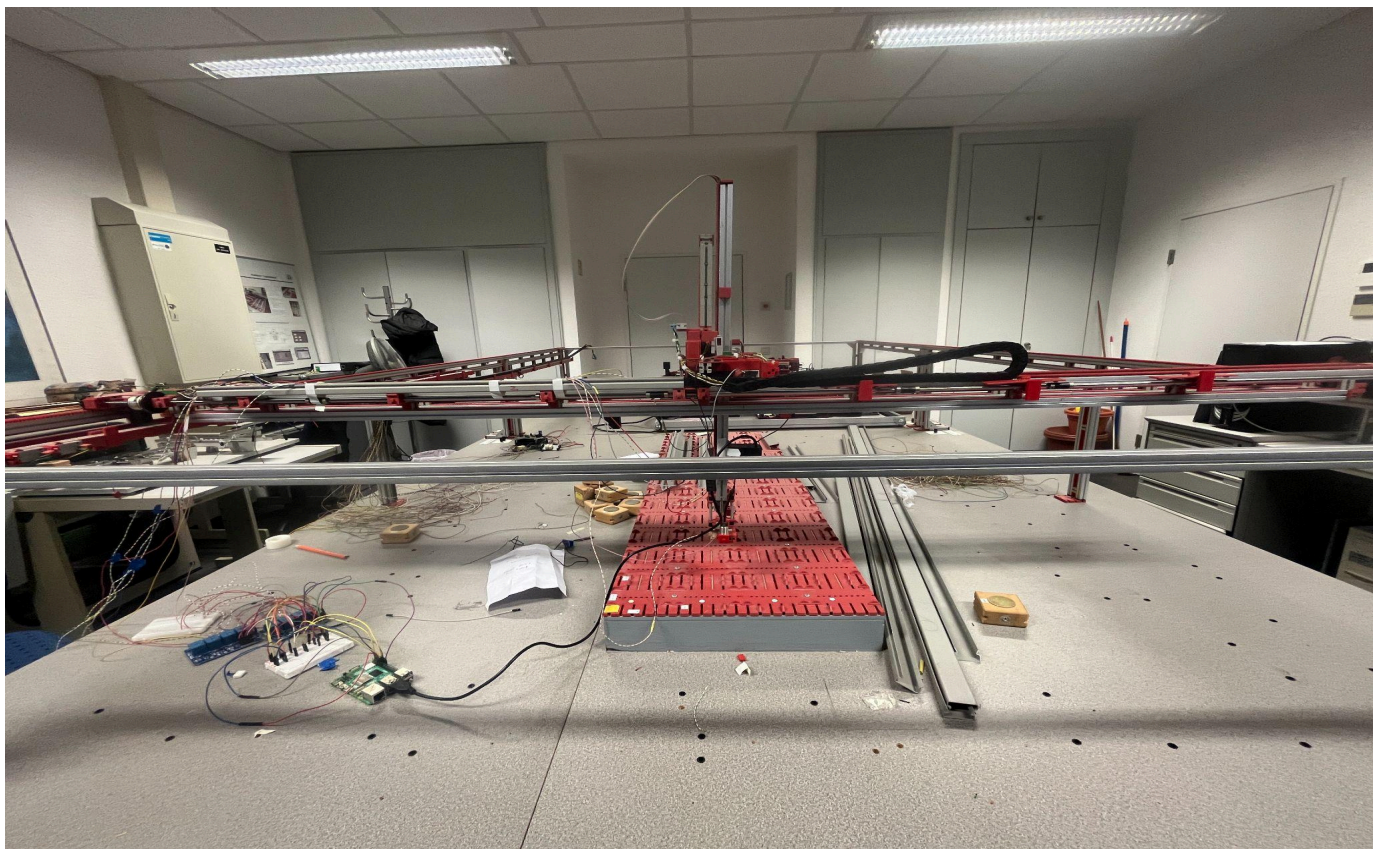
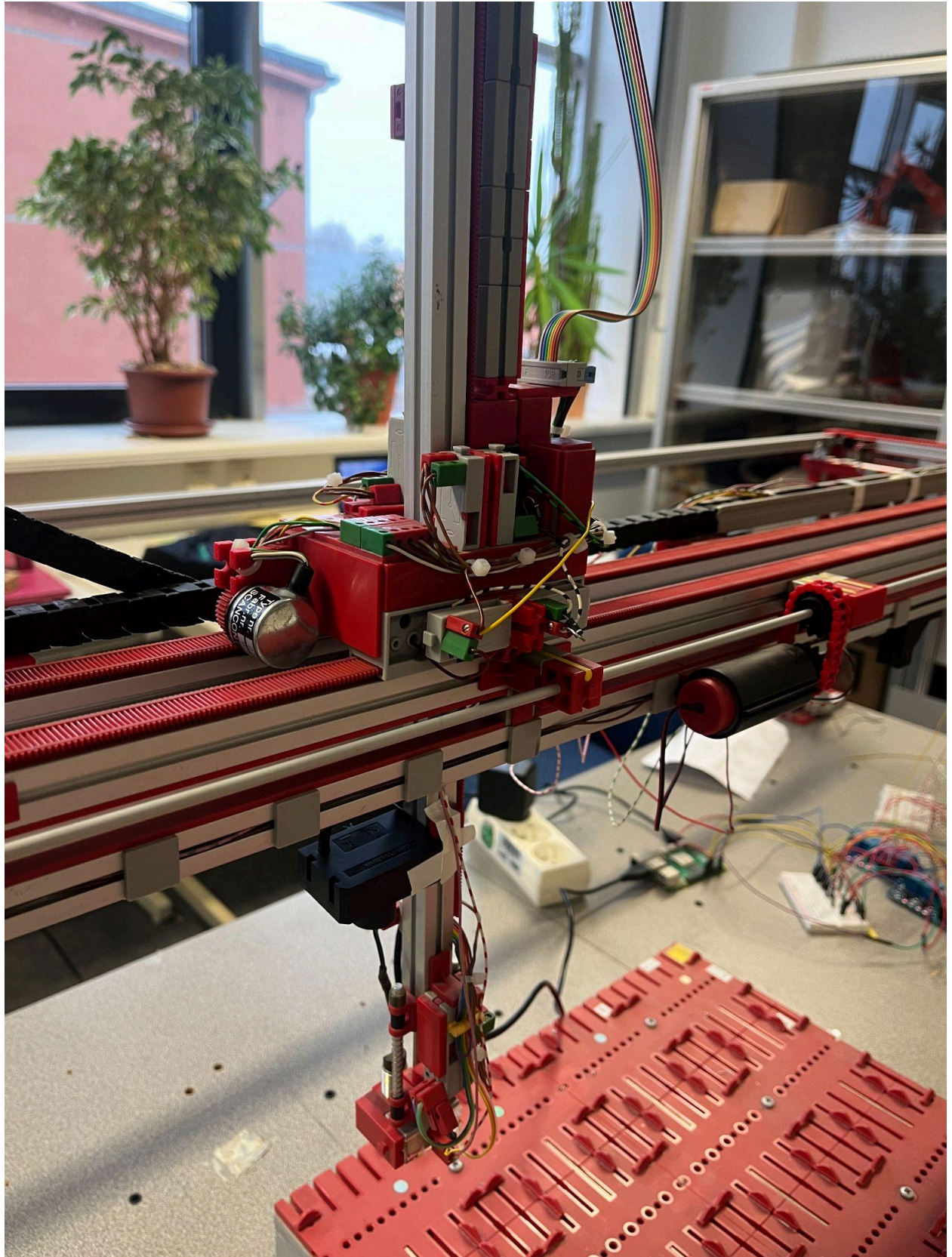


Computer Vision-Based Pick & Place Fischertechnik crane and Inventory Management





Computer Vision-Based Pick and Place Fischertechnik crane and Inventory management

1. Introduction

With the advancement of Industry 4.0, automation, intelligence, and digital integration have become essential in modern warehouse management systems. This project focuses on designing and implementing an automated warehouse solution that integrates **computer vision**, **microcontrollers**, and a **digital twin** to improve efficiency, accuracy, and scalability in logistics operations.

The system automates the process of **picking, storing, and dispatching orders** using a 3-axis gantry mechanism controlled by a Raspberry Pi and guided by object detection algorithms.

2. Problem Statement

Traditional warehouse operations rely heavily on manual labor, leading to:

- Human errors in picking and dispatching
- Increased operational time
- Limited real-time tracking and visibility

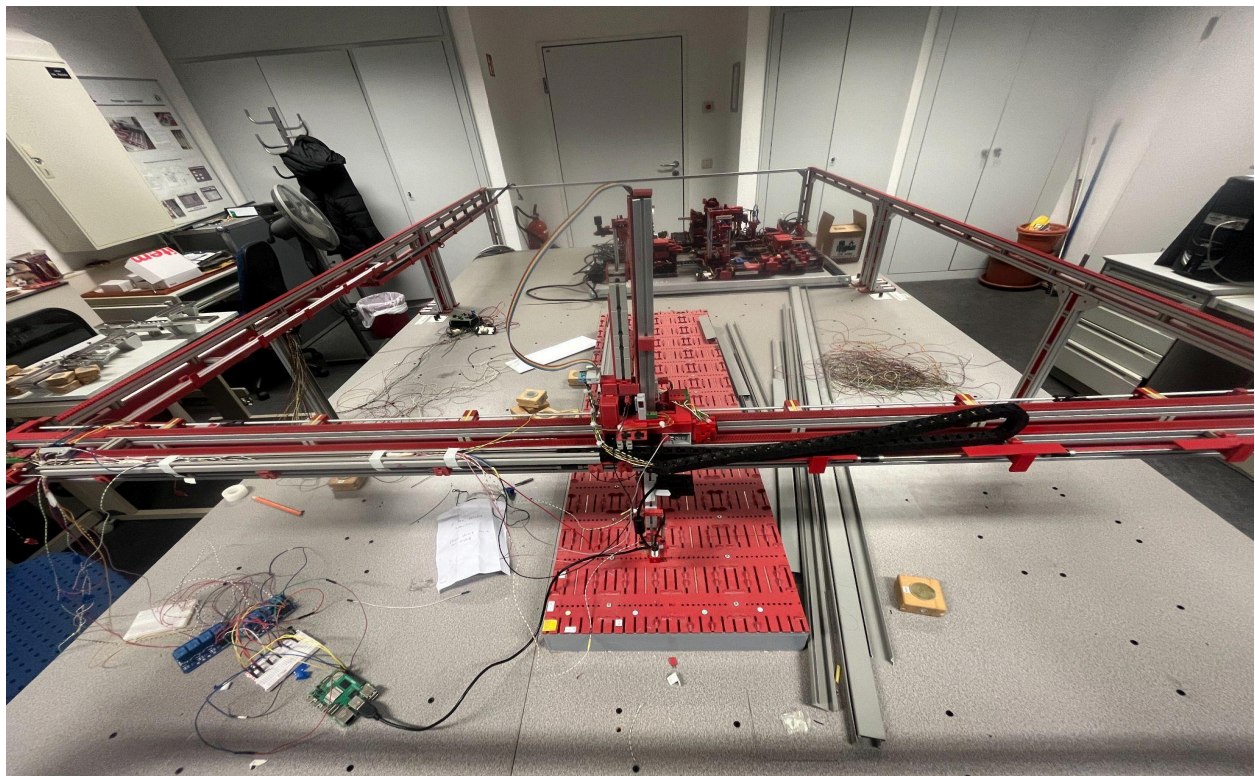
Objectives

- Automate pick-and-place operations using microcontrollers
- Integrate computer vision for object detection and identification
- Improve accuracy and efficiency in warehouse logistics
- Enable real-time order tracking through a user interface

3. System Concept

The core concept of this project is to combine **physical automation** with **intelligent perception** and **digital simulation**.

The system uses a camera-based object detection model to identify items in the warehouse. Based on the detected object's position, a Raspberry Pi controls DC motors via relays to move a gantry crane along X, Y, and Z axes. An electromagnet is used to pick and place objects. A digital twin mirrors the real system for testing and validation.



4. Concept Diagram (Logical View)

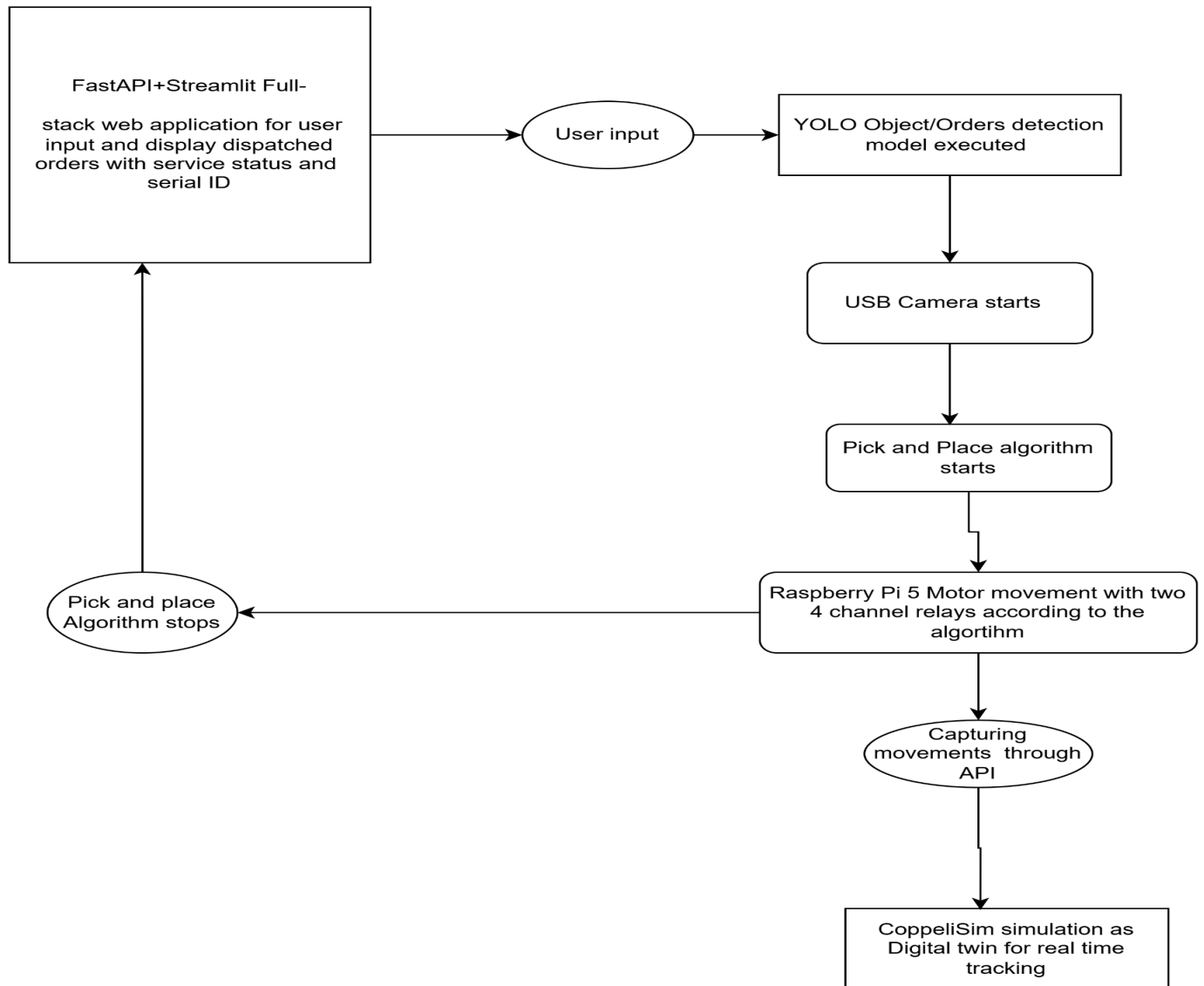


Fig.1 Concept Diagram

4. Hardware Used

- **Raspberry Pi 5** with 4-channel relay modules
 - **Electromagnet** for pick-and-place operations
 - **DC Motors** for X, Y, and Z axis movement
 - **USB Camera Module** for object detection
 - **12V Power Supply** for motor operation
-

5. Hardware Setup

- Raspberry Pi controls two 4-channel relay boards using digital GPIO pins
 - Each DC motor uses two relays for forward and reverse motion
 - Relays switch a 12V supply to drive the motors
 - Electrical isolation ensures protection from high motor currents
 - The electromagnet is mounted as an end-effector on the Z-axis
-

6. Digital Twin Setup

A digital twin of the system is developed using **CoppeliaSim** to replicate real-world behavior.

- 3-axis gantry system modeled using prismatic joints
 - Realistic motion constraints and limits applied
 - Electromagnet modeled for pick-and-place actions
 - Enables testing, debugging, and validation before physical deployment
-

7. Object Detection and System Integration

- Objects were labeled and a custom dataset was created
- Object detection model trained using **Ultralytics YOLO (yolo11s.pt)**
- Algorithm developed to:
 - Load the trained model
 - Perform inference on camera images
 - Draw bounding boxes with confidence scores
- Detected object coordinates are mapped to gantry movements
- Fully autonomous pick-and-place operation achieved

8. Model Evaluation and Improvement

Model performance was evaluated using validation datasets.

Metric	Value
mAP@50	50%
mAP@50–90	51%
Precision	53%
Recall	55%
Inference	5.0 s

Observations

- Moderate accuracy due to limited training images
- Inference speed is acceptable
- PyTorch model converted to **ONNX**, resulting in improved FPS

Future Improvements

- Train model with larger and more diverse datasets
- Improve detection accuracy and robustness

9. User Interface

- Backend developed using **FastAPI**
- Frontend created with **Streamlit**
- Features include:
 - Real-time tracking of dispatched orders
 - System status monitoring
 - User-friendly interaction with the automated system

10. Tools and Technologies Used

Software Tools

- Python
- Ultralytics YOLO
- OpenCV
- FastAPI
- Streamlit
- Coppeliasim
- ONNX

Hardware Tools

- Raspberry Pi 5
- Relay modules
- DC motors
- Camera module
- Electromagnet

11. Future Scope

- Integrate EasyOCR for serial number extraction
- Fully integrate digital twin with live system data

12. Conclusion

This project demonstrates a practical Industry 4.0 solution by combining automation, computer vision, and digital twins. The proposed system improves warehouse efficiency, reduces human error, and provides a foundation for intelligent, scalable logistics automation.